

DES MOINES, IA, USA

WMO: 725460

Lat: 41.534N

Lon: 93.653W

Elev: 292

StdP: 97.87

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14933

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.2	-17.5	-24.9	0.4	-19.6	-22.5	0.5	-17.1	13.4	-6.8	12.1	-5.6	4.2	310	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	33.9	24.5	32.3	23.9	30.8	23.1	26.1	31.7	25.2	30.5	24.2	29.1	5.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.4	20.1	29.8	23.5	19.0	28.9	22.6	18.0	27.8	82.5	31.8	78.5	30.4	74.7	29.3	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	10.1	8.8	-24.1	36.4	3.1	1.8	-26.3	37.6	-28.1	38.7	-29.8	39.7	-32.1	40.9		
(5)			-24.4	27.7	3.0	0.8	-26.6	28.3	-28.3	28.7	-30.0	29.2	-32.1	29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-5.2	-2.8	4.2	11.0	17.0	22.5	24.8	23.6	19.4	12.2	4.6	-2.2		
	DBStd	11.73	6.98	6.98	6.75	5.51	4.66	3.50	3.11	2.99	4.45	5.29	5.87	6.42		
	HDD10.0	1689	472	359	204	51	4	0	0	0	1	38	180	381		
	HDD18.3	3369	730	591	439	226	82	8	1	2	39	201	413	638		
	CDD10.0	1987	0	2	23	83	220	374	458	421	284	105	17	1		
	CDD18.3	626	0	0	1	7	41	131	200	164	73	10	0	0		
	CDH23.3	5673	0	0	10	78	344	1167	1974	1392	620	85	2	0		
	CDH26.7	2032	0	0	1	16	89	412	802	494	202	17	0	0		
(14) Wind	WSAvg	4.3	4.7	4.7	4.9	5.0	4.5	4.0	3.6	3.4	3.8	4.4	4.7	4.5		
(15) Precipitation	PrecAvg	879	25	31	56	96	111	126	99	107	84	65	46	35		
	PrecMax	1419	109	73	148	197	253	342	248	348	242	166	166	138		
	PrecMin	562	3	3	4	6	36	26	1	6	16	7	1	0		
	PrecStd	184	18	18	34	43	49	76	59	77	52	41	33	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.3	17.3	24.7	29.0	31.8	34.1	36.9	35.4	33.3	28.9	23.0	15.5		
		MCWB	7.5	10.6	15.7	18.0	21.9	24.0	25.2	24.2	22.9	19.8	14.9	11.4		
	2%	DB	9.3	13.5	20.9	25.6	29.0	32.3	34.4	33.0	31.1	25.7	19.0	11.8		
		MCWB	5.4	8.6	13.5	16.4	20.3	23.5	25.2	24.6	22.3	17.9	13.2	8.5		
	5%	DB	6.4	9.8	17.6	22.9	27.0	30.8	32.6	31.1	29.0	23.1	15.8	8.9		
		MCWB	3.5	6.2	12.2	14.6	18.9	22.6	24.8	23.6	21.4	16.7	11.2	5.8		
	10%	DB	3.7	6.7	14.5	20.3	25.0	29.2	30.8	29.4	27.0	20.6	13.4	6.3		
		MCWB	1.6	3.8	9.5	13.1	17.8	21.7	23.8	22.7	20.1	15.4	9.2	3.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.4	12.5	17.5	19.8	23.6	26.2	27.5	26.8	24.9	22.0	16.3	13.0		
		MCDB	11.7	16.1	22.7	26.4	29.8	32.3	33.8	32.3	31.1	26.2	20.4	14.7		
	2%	WB	5.6	8.9	15.2	17.6	21.8	25.0	26.5	25.6	23.6	19.8	14.1	9.3		
		MCDB	8.8	12.8	19.1	23.6	27.5	30.7	32.3	30.9	28.9	23.3	18.1	10.7		
	5%	WB	3.7	6.2	12.7	15.8	20.4	23.9	25.6	24.6	22.6	18.0	11.8	5.8		
		MCDB	6.4	9.8	17.3	21.4	25.3	29.1	30.9	29.5	27.4	21.6	15.4	8.7		
	10%	WB	1.7	3.8	9.8	14.0	19.0	22.8	24.7	23.7	21.4	15.9	9.5	3.7		
		MCDB	3.5	6.5	14.4	19.0	23.6	27.6	29.6	28.1	25.5	19.9	12.9	6.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	9.1	10.1	11.1	10.4	9.9	9.9	9.8	11.1	10.4	9.4	8.4		
		MCDBR	12.2	12.9	14.6	15.0	12.5	11.5	11.4	11.6	12.5	13.2	13.3	11.7		
	5% WB	MCWBR	9.1	8.9	8.6	8.3	6.2	5.5	4.8	5.2	5.8	7.1	8.4	8.6		
		MCWBR	11.5	12.4	12.6	13.0	11.0	10.3	10.3	10.1	10.8	10.8	11.9	10.6		
(40) Clear-Sky Solar Irradiance	taub		0.290	0.319	0.331	0.388	0.414	0.425	0.440	0.424	0.388	0.344	0.301	0.291		
	taud		2.420	2.341	2.401	2.259	2.255	2.270	2.215	2.274	2.327	2.445	2.504	2.438		
	Ebn at Noon		865	886	919	882	864	852	834	837	844	848	849	832		
	Edh at Noon		78	99	105	131	135	133	139	128	113	89	72	71		
(44) All-Sky Solar Radiation	RadAvg		1.93	2.93	3.79	4.75	5.61	6.24	6.29	5.46	4.59	3.09	2.10	1.62		
	RadStd		0.18	0.19	0.36	0.36	0.45	0.60	0.47	0.35	0.34	0.36	0.18	0.15		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page